there exist $C = C(n,\beta) > 0$ *and* $\lambda = \lambda(n,\beta) > 0$*, such that with probability at least* $1 - 2n^2 d^{-1/64}$*,*

$$\left|\langle x_i(t), x_j(t)\rangle - \gamma_\beta(t)\right| \leqslant \min\left\{2\cdot c(\beta)^{nt}\sqrt{\frac{\log d}{d}}, Ce^{-\lambda t}\right\} \tag{6.11}$$

holds for any $i \neq j$ *and* $t \geqslant 0$*, where* $c(\beta) = e^{10\max\{1,\beta\}}$*, and* γ_β *is the unique solution to* (6.9)*.*

Since the proof is rather lengthy, we defer it to Appendix D. It relies on combining the stability of the flow with respect to the initial data (entailed by the Lipschitz nature of the vector field) with concentration of measure. An analogous statement also holds for (USA), and more details can be found in Remark D.1, whereas the explicit values of C and λ can be found in (D.15). The upper bound in (6.11) is of interest in regimes where d and/or t are sufficiently large as the error in (6.11) is trivially bounded by 2.

Proof of Theorem 6.8. We split the proof in two parts. We focus on proving the result for the dynamics (SA), since the same arguments readily apply to the dynamics (USA).

Part 1. The angle $\theta_\beta(t)$*.* We first show there exists $\theta \in C^0(\mathbb{R}_{\geqslant 0};\mathbb{T})$ such that $\theta(t) = \angle(x_i(t), x_j(t))$ for any distinct $(i,j) \in [n]^2$ and $t \geqslant 0$. Since the initial tokens are orthogonal (and thus $d \geqslant n$), we may consider an orthonormal basis $(e_1,\dots,e_d)$ of $\mathbb{R}^d$ such that $x_i(0) = e_i$ for $i \in [n]$. Let $\pi : [d] \to [d]$ be a permutation. By decomposing any $x \in \mathbb{S}^{d-1}$ in this basis, we define $P_\pi : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ as

$$P_\pi\left(\sum_{i=1}^n a_i e_i\right) = \sum_{i=1}^n a_i e_{\pi(i)}.$$

Setting $y_i(t) = P_\pi(x_i(t))$ for $i \in [n]$, we see that $y_i(t)$ solves (SA) with initial condition $y_i(0) = P_\pi(x_i(0))$. But $(x_{\pi(1)}(t),\dots,x_{\pi(n)}(t))$ is a solution of (SA) by permutation equivariance, and it has the same initial condition since $P_\pi(x_i(0)) = x_{\pi(i)}(0)$. Consequently, we deduce that $P_\pi(x_i(t)) = x_{\pi(i)}(t)$ for any $t \geqslant 0$ and any $i \in [d]$. Hence

$$\langle x_i(t), x_j(t)\rangle = \langle P_\pi(x_i(t)), P_\pi(x_j(t))\rangle = \langle x_{\pi(i)}(t), x_{\pi(j)}(t)\rangle$$

which concludes the proof.

Part 2. The curve $\gamma_\beta(t)$*.* By virtue of the orthogonality assumption we have $\gamma_\beta(0) = \cos(\theta_\beta(0)) = 0$. To prove that $\gamma_\beta(t)$ satisfies (6.9) for the case of (SA), recall that

$$\mathbf{P}^\perp_{x_i(t)}(x_j(t)) = x_j(t) - \langle x_i(t), x_j(t)\rangle x_i(t).$$

Then for $k \neq i$,

$$\begin{aligned}\dot\gamma_\beta(t) &= 2\langle \dot x_i(t), x_k(t)\rangle \\ &= 2\sum_{j=1}^n \left(\frac{e^{\beta\langle x_i(t),x_j(t)\rangle}}{\sum_{\ell=1}^n e^{\beta\langle x_i(t),x_\ell(t)\rangle}}\right)\left(\langle x_j(t), x_k(t)\rangle - \langle x_i(t), x_j(t)\rangle\langle x_i(t), x_k(t)\rangle\right).\end{aligned}$$

Since the denominator in the above expression is equal to $(n-1)e^{\beta\gamma_\beta(t)} + e^\beta$, we end up with

$$\begin{aligned}\dot{\gamma}_\beta(t) &= \frac{2e^{\beta\gamma_\beta(t)}}{(n-1)e^{\beta\gamma_\beta(t)} + e^\beta} \sum_{j=1}^{n} \Big(\langle x_j(t), x_k(t)\rangle - \langle x_i(t), x_j(t)\rangle\langle x_i(t), x_k(t)\rangle \Big) \\ &= \frac{2e^{\beta\gamma_\beta(t)}}{(n-1)e^{\beta\gamma_\beta(t)} + e^\beta}(1-\gamma_\beta(t)^2 + (n-2)(\gamma_\beta(t) - \gamma_\beta(t)^2)),\end{aligned}$$

as desired. □

6.3. **Metastability and a phase transition.** An interesting byproduct of Theorem 6.8 and Theorem 6.9 is the fact that they provide an accurate approximation of the exact *phase transition curve* delimiting the clustering and non-clustering regimes, in terms of t and β. To be more precise, given an initial sequence $(x_i(0))_{i\in[n]} \in (\mathbb{S}^{d-1})^n$ of random points distributed independently according to the uniform distribution on $\mathbb{S}^{d-1}$, and for any fixed $0 < \delta \ll 1$, we define the phase transition curve as the boundary

$$\Gamma_{d,\delta} = \partial \left\{ t, \beta \geqslant 0 \colon t = \arg\inf_{s\geqslant 0} \Big(\mathbb{P}(\langle x_1(s), x_2(s)\rangle \geqslant 1-\delta) = 1 - 2n^2 d^{-\frac{1}{64}} \Big) \right\}$$

where $(x_i(\cdot))_{i\in[n]}$ denotes the solution to the corresponding Cauchy problem for (SA). (Here the choice of the first two particles instead of a random distinct pair is justified due to permutation equivariance.) Theorem 6.9 then gives the intuition that over compact subsets of $(\mathbb{R}_{\geqslant 0})^2$, $\Gamma_{d,\delta}$ should be well-approximated by

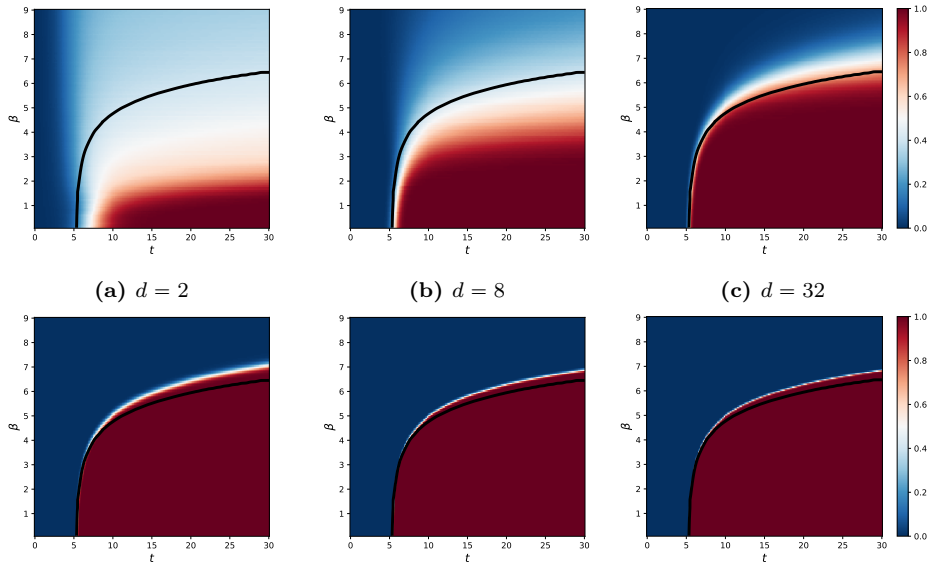
$$\Gamma_{\infty,\delta} = \Big\{ t, \beta \geqslant 0 \colon \gamma_\beta(t) = 1-\delta \Big\}. \tag{6.12}$$

This is clearly seen in Figure 3, along with the fact that the resolution of this approximation increases with $d \to +\infty$.

Figure 3 appears to contain more information than what we may gather from Theorem 6.3, Theorem 6.8 and Theorem 6.9. In particular, for small d, we see the appearance of a zone (white/light blue in Figure 3) of parameters (t, β) for which the probability of particles being clustered is positive, but not close to one. A careful inspection of this region reveals that points are grouped in a finite number of clusters; see Figure 4. The presence of such a zone indicates the emergence of a long-time *metastable* state where points are clustered into several groups but eventually relax to a single cluster in long-time. This two-time-scale phenomenon is illustrated in Figure 4 and prompts us to formulate the following question.

Problem 1. *Do the dynamics enter a transient metastable state, in the sense that for $\beta \gg 1$, all particles stay in the vicinity of $m < n$ clusters for long periods of time, before they all collapse to the final cluster $\{x^*\}$?*

There have been important steps towards a systematic theory of metastability for gradient flows, with applications to nonlinear parabolic equations—typically reaction-diffusion equations such as the Allen-Cahn or Cahn-Hilliard equations [OR07, KO02]—. While these tools to not readily apply to the current setup, they form an important starting point to answer this question.



(a) $d = 2$ **(b)** $d = 8$ **(c)** $d = 32$

(d) $d = 128$ **(e)** $d = 512$ **(f)** $d = 1024$

Figure 3. Plots of the probability that randomly initialized particles following (SA) cluster to a single point as a function of t and β: we graph the function $(t, \beta) \mapsto \mathbb{P}_{(x_1(0),\ldots,x_n(0))\sim\sigma_d}(\{\langle x_1(t), x_2(t)\rangle \geqslant 1-\delta\})$, which is equal to $(t, \beta) \mapsto \mathbb{P}_{(x_1(0),\ldots,x_n(0))\sim\sigma_d, i\neq j \text{ fixed}}(\{\langle x_1(t), x_2(t)\rangle \geqslant 1-\delta\})$ by permutation equivariance. We compute this function by generating the average of the histogram of $\{\langle x_i(t), x_j(t)\rangle \geqslant 1-\delta \colon (i,j) \in [n]^2, i \neq j\}$ over 2^{10} different realizations of initial sequences. Here, $\delta = 10^{-3}$, $n = 32$, while d varies. We see that the curve $\Gamma_{\infty,\delta}$ defined in (6.12) approximates the actual phase transition with increasing accuracy as d grows, as implied by Theorem 6.9.

Finally, one may naturally ask whether the clustering and phase diagram conclusions persist when the parameter matrices (Q, K, V) are significantly more general: some illustrations[11] are given in Figure 5.

Problem 2. *Can the conclusions of Theorem 6.8–Theorem 6.9 be generalized to the case of random matrices (Q, K, V)?*

Part 3. Further questions

We conclude this manuscript by discussing several avenues of research that can lead to a finer understanding of the clustering phenomenon and generalizations of our results, and which, we believe, are of independent mathematical interest. Specifically,

- In Section 7, we zero in on the special case $d = 2$, where we make a link with the celebrated *Kuramoto model* when $\beta = 0$;

[11] See github.com/borjanG/2023-transformers-rotf for additional figures which indicate that this phenomenon appears to hold in even more generality.